

Other topics in Database Systems

- **Transaction Management**
- **Concurrency control**
- **Security, Backup and Recovery**

Transaction Management

What is a Transaction?

- Any action that reads from and/or writes to a database and may consist of
 - Simple SELECT statement to generate a list of table contents
 - A series of related UPDATE statements to change the values of attributes in various tables
 - A series of INSERT statements to add rows to one or more tables
 - A combination of SELECT, UPDATE, and INSERT statements
- A *logical* unit of work that must be either entirely completed or aborted
- Successful transaction changes the database from one *consistent* state to another
 - One in which all data integrity constraints are satisfied

Transaction Processing

- For example, a transaction may involve
 - The creation of a new invoice
 - Insertion of a row in a table
 - Decreasing the quantity on hand by 1
 - Updating the customer balance
 - Creating a new account transaction row
- If the system fails between the first and last step, the database will no longer be in a consistent state

Transaction- Concurrency in Database

- A transaction usually means an interaction among 2 or more parties in the conduct of business e.g., a customer withdrawing money from the bank account, the purchase of a car or booking a flight.
- In a **DBMS**, this type of transaction may require *several database operations to take place*.
- To withdraw money, the operations might be:
 - Check the account balance to make sure the withdrawal amount is available
 - Insert a new record to the AccountHistory table
 - Update the account balance in the Accounts table

Cont...

- In a database system, one transaction can involve *any number of reads from and writes to the database*. Any one of the operations e.g. check the account balance does not make sense to the end user if taken on its own.
- A collection of database operations taken together is a single unit from the point of view of the database user.
- A definition of **transaction**, in database terms: – **a transaction is a sequence of operations performed as a single, logical unit of work.**
- The operations performed as part of a transaction make changes to the data i.e. they are **data modification operations**

Cont...

- At any point in time, there may be many transactions happening in a system. These are concurrent transactions
- A DBMS should be able to handle *shared update* of the database, which means concurrent transactions updating the same data
- The **key feature** of a transaction is that **all the operations must succeed**.
- If any one of the operations fails, then the transaction must fail – it must undo any of the operations that did succeed.

Cont...

- For a logical unit of work to be considered a transaction, it should have certain properties.
- These properties help to ensure the integrity of data in a database. They are:
 - *Atomicity*
 - *Consistency*
 - *Isolation*
 - *Durability*
- These are known as the ACID properties of a transaction (from the first letters).

Transaction Properties

- **Atomicity**

- Requires that *all* operations (SQL requests) of a transaction be completed; if not, then the transaction is aborted
- A transaction is treated as a single, indivisible, logical unit of work

- **Consistency/Serializability**

- Ensures that the concurrent execution of several transactions yields consistent results
- Multiple, concurrent transactions appear as if they executed in serial order (one after another)

Transaction Properties

- **Isolation**

- Data used during execution of a transaction cannot be used by second transaction until first one is completed

- **Durability**

- Indicates permanence of database's consistent state
- When a transaction is complete, the database reaches a consistent state. That state can not be lost even if the system fails

Transaction Management with SQL

- ANSI has defined standards that govern SQL database transactions
- Transaction support is provided by two SQL statements: COMMIT and ROLLBACK

Transaction Management with SQL

1. A **COMMIT** statement is reached- all changes are permanently recorded within the database
2. A **ROLLBACK** is reached – all changes are aborted and the database is restored to a previous consistent state (rolled back or undone)

The Transaction Log

- Keeps track of all transactions that update the database. It contains:
 - A record for the beginning of transaction
 - For each transaction component (SQL statement)
 - Type of operation being performed (update, delete, insert)
 - Names of objects affected by the transaction (the name of the table)
 - “Before” and “after” values for updated fields
 - Pointers to previous and next transaction log entries for the same transaction
 - The ending (COMMIT) of the transaction
- Increases processing overhead but the ability to restore a corrupted database is worth the price

Concurrency Control

Concurrency Control (CC)

- The coordination of the simultaneous execution of transactions in a multiprocessing database is known as *concurrency control*
- The objective of concurrency control is to ensure the serializability of transactions in a multiuser database environment
- Importance → simultaneous execution of transactions over a shared database can create several data integrity and consistency problems
 - The three main problems are:
 - *lost updates*
 - *uncommitted data*
 - *inconsistent retrievals*

Problems

- **Uncommitted data** occurs when two transactions execute concurrently and the first is rolled back after the second has already accessed the uncommitted data.
 - This violates the isolation property of transactions
- **Inconsistent Retrieval Problem**: occur when a transaction calculates some aggregate functions over a set of data while transactions are updating the data
 - Some data may be read after they are changed and some before they are changed yielding inconsistent results
- **Lost update**: occurs when update by one transaction is overwritten by the other

Concurrency Control with Locking Methods

- Guarantees exclusive use of a data item to a current transaction
 - Transaction acquires a lock prior to data access; the lock is released when the transaction is complete
 - Required to prevent another transaction from reading inconsistent data
- **Lock Granularity**: Indicates the level of lock use (amount of data to be locked)
- Locking can take place at the following levels:
 - Database-level lock: Entire database is locked
 - Table-level lock: Entire table is locked
 - Field-level lock : one specific field/attribute is locked.

Deadlocks

- Condition that occurs when two transactions wait for each other to unlock data
 - T1 needs data items X and Y; T needs Y and X
 - Each gets the its first piece of data but then waits to get the second (which the other transaction is already holding) – *deadly embrace*
- Possible only if one of the transactions wants to obtain an exclusive lock on a data item
 - No deadlock condition can exist among *shared* locks
- Control through
 - *Prevention*
 - *Detection*
 - *Avoidance*

Database Security Backup and Recovery

Database Security

- A database should be properly secured using appropriate controls.
- Database security encompasses hardware, software, people and data
- Database security and integrity is about protecting the database from being inconsistent and being disrupted. We can also call it database misuse.
- Database misuse could be Intentional or accidental, where accidental misuse is easier to cope with than intentional misuse.

Cont...

- Accidental inconsistency could occur due to:
 - System crash during transaction processing
 - Anomalies due to concurrent access
 - Anomalies due to redundancy
- Like wise, even though there are various threats that could be categorized in this group, intentional misuse could be:
 - Unauthorized reading of data
 - Unauthorized modification of data or
 - Unauthorized destruction of data

Cont...

- Most systems implement good **Database Integrity** to protect the system from accidental misuse while there are many computer based measures to protect the system from intentional misuse, which is termed as **Database Security** measures.
- Database security is considered in relation to the following situations:
 - *Theft and fraud*
 - *Loss of confidentiality (secrecy)*
 - *Loss of privacy*
 - *Loss of integrity*
 - *Loss of availability*

Cont...

- Security Issues and general considerations
 - Legal, ethical and social issues regarding the right to access information
 - Policy issues regarding privacy of individual level at enterprise and national level
 - Operational consideration on the techniques used (password, etc)
 - System level security including operating system and hardware control
 - Security levels and security policies in enterprise level

Cont...

- Thus **database security – intimidation (threat)**
- **Threat** – any situation or event, whether intentional or accidental, that may adversely affect a system and consequently the organization
 - A threat may be caused by a situation or event involving a person, action, or circumstance that is likely to bring harm to an organization

Cont...

- The harm to an organization may be *tangible* or *intangible*
 - *Tangible* – loss of hardware, software, or data
 - *Intangible* – loss of credibility or client confidence
- Examples of threats (intimidations):
 - Using another persons' means of access
 - Unauthorized amendment/modification or copying of data
 - Program alteration
 - Inadequate policies and procedures that allow a mix of confidential and normal out put
 - Illegal entry by hacker
 - Theft of data, programs, and equipment

Cont...

- Failure of security mechanisms, giving greater access than normal
- Data corruption owing to power loss or surge
- Fire (electrical fault, lightning strike, arson), flood, bomb
- Physical damage to equipment
- Breaking cables or disconnection of cables
- Introduction of viruses

Security Mechanisms

- Discretionary Security Mechanisms
 - To grant privileges to users, including the capability to access specific data files, records, or fields in a specific mode (such as read, insert, delete, or update).
- Mandatory Security Mechanisms
 - To enforce multilevel security by classifying the data and users into various security classes (or levels) and then implementing the appropriate security policy of the organization.

Cont...

- *Levels of Security Measures*

- Security measures can be implemented at several levels and for different components of the system. These levels are:
 - **Physical Level:** concerned with securing the site containing the computer system should be physically secured.
 - The backup systems should also be physically protected from access except for authorized users.
 - **Human Level:** concerned with authorization of database users for access the content at different levels and privileges.

Cont...

- **Operating System:** concerned with the weakness and strength of the operating system security on data files.
 - Weakness may serve as a means of unauthorized access to the database.
 - This also includes protection of data in primary and secondary memory from unauthorized access.
- **Database System:** concerned with data access limit enforced by the database system.
 - Access limit like password, isolated transaction and etc.
- Even though we can have different levels of security and authorization on data objects and users, *who access which data is a policy matter rather than technical.*

Cont...

- **Computer Based Countermeasures**
 - The types of countermeasure to threats on computer systems range from **physical controls to administrative procedures**
- Despite the range of computer-based controls that are available, it is worth noting that, generally, *the security of a DBMS is only as good as that of the operating system*, owing to their close association

Cont...

- The following are computer-based security controls for a multi-user environment:
- **Views**
 - A view is the dynamic result of one or more relational operations operation on the base relations to produce another relation
 - A view is a virtual relation that does not actually exist in the database, but is produced upon request by a particular user
 - The view mechanism provides a powerful and flexible security mechanism by hiding parts of the database from certain users
 - Using a view is more restrictive than simply having certain privileges granted to a user on the base relation(s)

Cont...

- **Integrity**

- Integrity constraints contribute to maintaining a secure database system by preventing data from becoming invalid and hence giving misleading or incorrect results
 - Domain Integrity
 - Entity integrity
 - Referential integrity
 - Key constraints

Cont...

- **Backup and recovery**

- Backup is the process of periodically taking a copy of the database and log file (and possibly programs) on to offline storage media
- A DBMS should provide backup facilities to assist with the recovery of a database following failure
- Database recovery is the process of restoring the database to a correct state in the event of a failure
- Journaling is the process of keeping and maintaining a log file (or journal) of all changes made to the database to enable recovery to be undertaken effectively in the event of a failure

Cont...

- The advantage of journaling is that, in the event of a failure, the database can be recovered to its last known consistent state using a backup copy of the database and the information contained in the log file
- If no journaling is enabled on a failed system, the only means of recovery is to restore the database using the latest backup version of the database
- However, without a log file, any changes made after the last backup to the database will be lost

Cont...

- **Encryption**

- The encoding of the data by a special algorithm that renders the data unreadable by any program without the decryption key
- If a database system holds particularly sensitive data, it may be deemed necessary to encode it as a precaution against possible external threats or attempts to access it
- The DBMS can access data after decoding it, although there is a degradation in performance because of the time taken to decode it
- Encryption also protects data transmitted over communication lines

Cont...

- **Authorization**

- All users of the database will have different access levels and permission for different data objects, and authentication is the process of checking whether the user is the one with the privilege for the access level.
- Is the process of checking the users are who they say they are.
- Each user is given a unique identifier, which is used by the operating system to determine who they are
- Thus the system will check whether the user with a specific username and password is trying to use the resource.

Cont...

- The granting of a right or privilege that enables a subject to have legitimate access to a system or a system's object
- Authorization controls can be built into the software, and govern not only what system or object a specified user can access, but also what the user may do with it
- Authorization controls are sometimes referred to as *access controls*
- The process of authorization involves authentication of *subjects* (i.e. a user or program) requesting access to *objects* (i.e. a database table, view, procedure, trigger, or any other object that can be created within the system)

Cont...

- Any database access request will have the following three major components
 - **Requested Operation:** what kind of operation is requested by a specific query?
 - **Requested Object:** on which resource or data of the database is the operation sought to be applied?
 - **Requesting User:** who is the user requesting the operation on the specified object?
- The database should be able to check for all the three components before processing any request.
- The checking is performed by the security subsystem of the DBMS.

Cont...

- There are different forms of user authorization on the resource of the database.
- These forms are privileges on what operations are allowed on a specific data object.
- **User authorization on the data/extension**
 - **Read Authorization:** the user with this privilege is allowed only to read the content of the data object.
 - **Insert Authorization:** the user with this privilege is allowed only to insert new records or items to the data object.

Cont...

- **Update Authorization:** users with this privilege are allowed to modify content of attributes but are not authorized to delete the records.
- **Delete Authorization:** users with this privilege are only allowed to delete a record and not anything else.
 - Different users, depending on the power of the user, can have one or the combination of the above forms of authorization on different data objects.

Cont...

- The major responsibilities of DBA in relation to authorization of users are:
 - ***Account Creation:*** involves creating different accounts for different **USERS** as well as **USER GROUPS**.
 - ***Security Level Assignment:*** involves in assigning different users at different categories of access levels.
 - ***Privilege Grant:*** involves giving different levels of privileges for different users and user groups.

Cont...

- ***Privilege Revocation:*** involves denying or canceling previously granted privileges for users due to various reasons.
- ***Account Deletion:*** involves in deleting an existing account of users or user groups. Is similar with denying all privileges of users on the database

Granting Privileges

- The owner account holder can pass privileges on any of the owned relations to other users by granting privileges to their accounts.
- In SQL the following types of privileges can be granted on each individual relation R:
 - SELECT
 - MODIFY (UPDATE, DELETE, INSERT)
 - INSERT and UPDATE - certain attributes

REVOKE & GRANT OPTION

- REVOKE
 - To cancel privileges
- GRANT option
 - If grant option is given then the account can also grant that privilege on the relation to other accounts.
 - Suppose that B is given the GRANT OPTION by A and that B then grants the privilege on R to a third account C, also with GRANT OPTION.
 - In this way, privileges on R can propagate to other accounts without the knowledge of the owner of R.
 - If the owner account A now revokes the privilege granted to B, all the privileges that B propagated based on that privilege should automatically be revoked by the system.

😊 It is Over 😊

- Finally:
 - About your Project
 - About the Final Exam
-

☺What To Submit☺

1. The Full Documentation according to the Guideline
2. A CD Containing the Database
 - You have to Detach the database from SQL and burn on a CD

The End